

Nmap - Cheat Sheet

A general network scanning workflow with nmap. Replace placeholders (<...>) with your own values.

Step 1 — Host Discovery

Find live hosts before deep-scanning:

```
nmap -sn <target-range>
```

Target formats: 192.168.1.1 (single host), 192.168.1.0/24 (subnet), 192.168.1.1-50 (range).

-Pn — skip ping, treat all hosts as up (use when ICMP is blocked).

Step 2 — Port Scan

Choose a scan type:

```
nmap -sS <target> # stealth SYN — recommended, requires root
nmap -sT <target> # TCP connect — no root needed, more noise
nmap -sU <target> # UDP — slow, combine with --top-ports
```

Flag	Name	Description
-sS	SYN Stealth	Half-open scan, never completes the handshake. Fast and stealthy. Requires root.
-sT	TCP Connect	Full 3-way handshake. No root needed but easily logged.
-sU	UDP Scan	Scans UDP ports. Slow — combine with --top-ports to limit scope.
-sA	ACK Scan	Maps firewall rules. Shows filtered vs. unfiltered, not open/closed.
-sN/-sF/-sX	Null/FIN/Xmas	No-flag, FIN-only, or FIN+PSH+URG. Evade some firewalls (UNIX targets only).
-p <ports>	Port Spec	e.g. -p 80, -p 1-1000, -p 22,445,3389
-p-	All Ports	Scan all 65,535 TCP ports.
-F	Fast	Top 100 most common ports only.
--top-ports N	Top N	Scan the N most commonly open ports.
-T<0-5>	Timing	T0=Paranoid (stealthiest) ... T3=Normal (default) ... T5=Insane (fastest/noisiest).

Step 3 — Version and OS Detection

```
nmap -sV <target> # service + version detection
nmap -O <target> # OS fingerprinting (requires root)
nmap -A <target> # aggressive: -sV + -O + scripts + traceroute
```

-n disables reverse DNS (speeds up scans). -v / -vv for more output detail.

Step 4 — NSE Scripts

```
nmap --script=default <target> # same as -sC
nmap --script=vuln <target> # known CVE checks
nmap --script=auth <target> # default/weak credential tests
```

Script	What it does
smb-vuln-ms17-010	EternalBlue (WannaCry) check on port 445
smb-enum-shares	List accessible SMB shares
smb-enum-users	Enumerate users via SMB
ldap-search	Query Active Directory over LDAP (port 389)
ftp-anon	Check for anonymous FTP login

Script	What it does
http-title	Grab HTTP page titles from web servers
ssh-brute	Brute-force SSH with a user/pass list

Step 5 — Save Output

```
nmap -oN <output.txt> # human-readable nmap -oX <output.xml> # XML – importable into Metasploit, etc. nmap -oA <basename> # save all three formats at once
```

Common Scan Combos

Goal	Command
Quick subnet sweep	<code>nmap -sn 192.168.1.0/24</code>
Stealth + version, save output	<code>nmap -sS -sV -T4 -oN &lt;output.txt> &lt;target></code>
Full aggressive scan (all ports)	<code>nmap -A -p- -T4 &lt;target></code>
EternalBlue check	<code>nmap -p 445 --script smb-vuln-ms17-010 &lt;target></code>
Full vulnerability scan	<code>nmap -sV --script vuln -p- &lt;target></code>
UDP top-20 ports	<code>nmap -sU --top-ports 20 &lt;target></code>

Key idea: always start broad (host discovery, top ports) and then narrow down. -A is great for CTFs and labs; use -sS with careful timing for stealthier engagements.